ProCo: Verifiable Abstention through Missing-Support Supervision

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METHODS • CONTROLLED STUDIES • VERIFIABLE EVIDENCE

Open-world reasoning needs evidence for all three outcomes. Missing-support supervision turns an unsupported answer into an object a symbolic verifier can inspect.

ProofWriter OWA • depth-3ext-NatLang → depth-5 • Qwen2.5-7B-Instruct • QLoRA, 4,096 questions, 3 seeds

01 Supervise the decision, expose the support, verify the evidence

A. Indexed theory and intervention

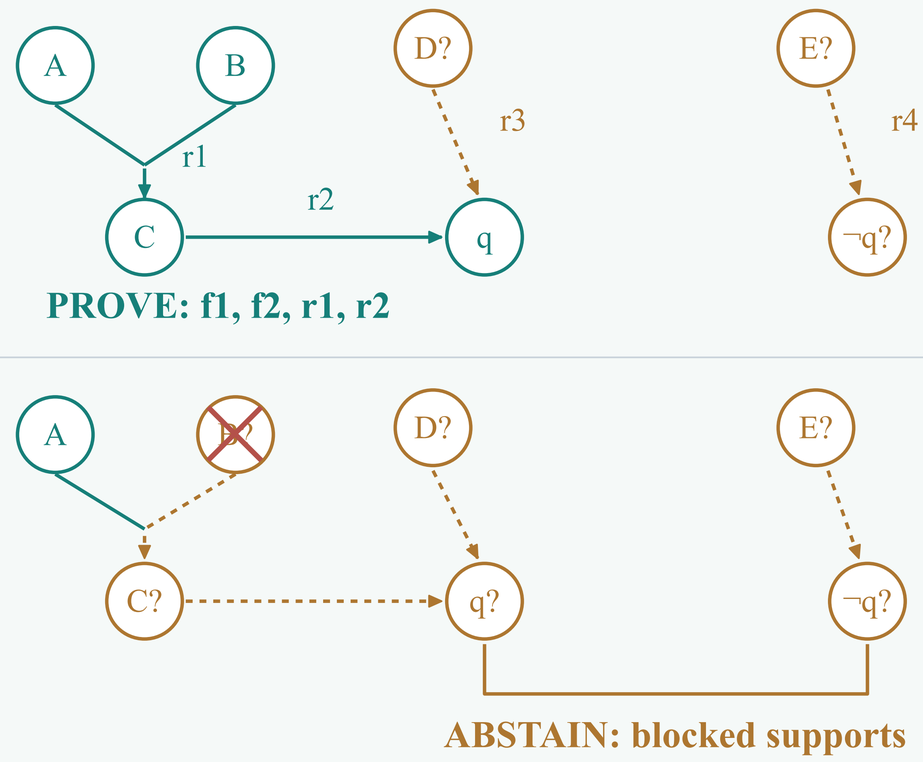
ID	Fact or rule
f1	**A**
f2	**B**
r1	**A AND B → C**
r2	**C → q**
r3	**D → q**
r4	**E → ¬q**

Original closure derives C and q. Delete f2, recompute closure, and retain the mutant only if neither q nor ¬q is derivable.

The causal edit is known

B is removed; q loses its supported chain.

B. Positive, opposite, and blocked support



Solid: derivable. Dashed: unavailable support.

C. Output contract

PROVE

Derivation of q, citing facts and rule applications.

REFUTE

Derivation of ¬q; explicit negative evidence.

ABSTAIN

A missing-support witness, checked against the theory.

Training target: (decision, evidence) • Joint correctness = correct label AND verifier-accepted evidence

Illustrative theory and proof objects; notation abstracts the serialized output grammar.

02 Three-seed main evidence

Joint (%)	In-domain	Depth 5
Proof-only	39.9 ± 0.6	**26.5 ± 1.1**
ProCo	75.0 ± 0.7	**55.0 ± 0.7**
Gain	+35.1 pp	**+28.5 pp**

Raw accuracy uses a different endpoint

Accuracy (%)	In-domain	Depth 5
Answer-only	88.9 ± 0.3	**80.0 ± 1.0**
Proof-only	87.9 ± 0.2	**76.1 ± 0.7**
ProCo	87.4 ± 0.3	**76.4 ± 0.6**

Answer-only remains a strong classifier. Since it emits no evidence, its zero joint score is definitional; proof-only is the substantive evidence baseline.

Each seed sees 4,096 training questions for one epoch. Fixed 4,000-example evaluation subsets separate training variation from test resampling.

03 Support-deletion stress test

Start from provable queries. Delete a necessary fact or rule, recompute the theory closure, and collect 4,000 gold-Unknown mutants.

Three seeds	Proof-only	ProCo
Pred. Unknown	2,412.0	**2,187.3**
Accepted witness	0	**2,017.7**
Joint (%)	0	**50.4 ± 1.0**

2,017.7 ± 40.6

verifier-accepted missing-support witnesses

ProCo abstains less often on these mutants yet produces many more accepted witnesses. The intervention probes evidence quality, beyond the frequency of Unknown labels.

The source of missing support is controlled. Remaining uncertainty concerns whether the generated witness satisfies the chosen symbolic verification rule.

04 What survives a stricter check

Gold-Unknown (%)	ID	Depth 5
Main accepted	77.7 ± 0.4	**68.6 ± 1.7**
Strict accepted	71.9 ± 0.5	**63.3 ± 1.5**
Rule strict	15.8 ± 1.7	**13.1 ± 0.8**
No-rule strict	87.1 ± 0.8	**81.7 ± 2.0**

Strict matching retains substantial evidence coverage. However, no-rule instances are common: 1,463/1,860 in-domain and 1,246/1,704 at depth 5.

Canonical form versus valid explanation

A blocked rule may admit several defensible missing-support explanations. Exact canonical matching is more restrictive than accepting equivalent witnesses.

Rule-based strict acceptance remains much lower. The current evidence supports a controlled symbolic result, with witness equivalence an unresolved verifier-design issue.

05 From a label to a reasoning object

Answer-only supervises True / False / Unknown. Proof-only adds structured reasoning but lacks explicit missing-support targets. ProCo assigns a concrete evidence regime to every decision: a proof, a refutation, or an abstention witness.

The gain concentrates in gold-Unknown cases: approximately 1,444 accepted cases in-domain and 1,169 under depth shift.

06 Data scale and scope

Full-data, one seed	Raw accuracy	Joint
In-domain	98.2%	**96.8%**
Depth 5	93.8%	**86.2%**

Larger-data results are a single-seed probe. Current validation is restricted to processed ProofWriter open-world theories with clean symbolic closure; transfer to unrestricted evidence spaces remains open.